Acquisition Investment Discipline and the Cross Section of Stock Returns

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Abstract

This paper studies the asset pricing implications of Acquisition Investment Discipline (AID), and its robustness in predicting returns in the cross-section of equities using the protocol proposed by Novy-Marx and Velikov (2023a). A value-weighted long/short trading strategy based on AID achieves an annualized gross (net) Sharpe ratio of 0.43 (0.35), and monthly average abnormal gross (net) return relative to the Fama and French (2015) five-factor model plus a momentum factor of 23 (16) bps/month with a t-statistic of 2.50 (1.80), respectively. Its gross monthly alpha relative to these six factors plus the six most closely related strategies from the factor zoo (Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing) is 22 bps/month with a t-statistic of 2.38.

1 Introduction

Asset prices reflect firms' production decisions and investment opportunities in equilibrium. The cross-section of stock returns exhibits systematic patterns related to corporate investment behavior, suggesting that firms' capital allocation decisions contain information about their risk exposures and expected returns. Despite extensive research on investment-based anomalies, significant gaps remain in understanding how specific aspects of firms' acquisition strategies relate to their cost of capital and risk premia.

We examine whether Acquisition Investment Discipline (AID) predicts cross-sectional variation in stock returns through the lens of production-based asset pricing. Firms exhibiting greater discipline in their acquisition investments face different marginal costs of adjustment and exposure to aggregate productivity shocks compared to firms with less disciplined acquisition strategies. This heterogeneity in acquisition behavior creates cross-sectional variation in risk exposures that rational investors price in equilibrium.

In a production-based economy, firms' optimal investment decisions depend on the interaction between productivity shocks, adjustment costs, and the stochastic discount factor Cochrane and Piazzesi (2005). The first-order conditions from firms' value maximization problems link expected returns directly to the marginal product of capital and the shadow value of investment constraints Liu et al. (2009). Acquisition Investment Discipline captures variation in these fundamental production characteristics, as disciplined acquirers face different investment frictions and productivity dynamics than undisciplined acquirers.

Firms with high acquisition discipline operate with lower adjustment costs and more flexible production technologies, enabling them to respond more efficiently to productivity shocks Zhang (2005). These firms exhibit lower systematic risk because their investment policies create valuable real options that hedge against adverse productivity realizations. The production function curvature and decreasing returns to scale imply that disciplined acquisition strategies reduce firms' exposure to aggregate investment shocks Balas and Novy-Marx (2019). Consequently, high-AID firms command lower risk premia in equilibrium as their cash flows covary less with the pricing kernel.

The cross-sectional pricing implications follow from the equivalence between production-based and consumption-based Euler equations in general equilibrium Cochrane (2011). Acquisition discipline proxies for firms' exposure to investment-specific technology shocks, which represent a priced source of systematic risk in production economies. Firms with low acquisition discipline face higher costs of adjusting their capital stock and greater exposure to aggregate productivity fluctuations Kogan and Papanikolaou (2013). These firms require higher expected returns to compensate investors for bearing their elevated systematic risk, generating the predicted negative relationship between AID and expected returns.

We find strong evidence that Acquisition Investment Discipline predicts cross-sectional stock returns. The value-weighted long/short trading strategy based on AID achieves an annualized gross Sharpe ratio of 0.43 and a monthly average excess return of 27 basis points with a t-statistic of 3.05. The strategy's performance remains robust across multiple factor model specifications, with the monthly alpha relative to the Fama-French six-factor model equaling 23 basis points with a t-statistic of 2.50.

The economic magnitude of the AID effect is substantial and persists after accounting for transaction costs. The net Sharpe ratio equals 0.35, placing the strategy in the top 7% of documented anomalies. Among the largest stocks—those with market capitalization above the 80th NYSE percentile—the AID strategy achieves an average return of 30 basis points per month with a t-statistic of 2.71. The strategy's alpha relative to the Fama-French five-factor model equals 25 basis points per month with a t-statistic of 2.20 in this large-cap subsample, demonstrating that the effect is not confined to small, illiquid stocks.

The AID signal provides incremental explanatory power beyond closely related investment-based anomalies. Controlling for six related predictors including Asset growth, Change in net financial assets, and Growth in book equity, the strategy maintains a significant alpha of 22 basis points per month with a t-statistic of 2.38. The Fama-MacBeth regression coefficient on AID remains statistically significant at -0.67 with a t-statistic of -1.82 when controlling for all six related anomalies simultaneously. These results confirm that AID captures a distinct dimension of cross-sectional return variation linked to firms' production characteristics.

Our study extends the production-based asset pricing literature by identifying acquisition discipline as a novel predictor of cross-sectional returns. While Cooper et al. (2008) document that asset growth predicts returns and Lyandres and Wahal (2019) examine post-acquisition performance, we provide the first systematic evidence linking acquisition investment discipline to risk premia through production-based channels. Our results complement Eisfeldt and Papanikolaou (2013) by showing that acquisition decisions reflect firms' exposure to organization capital risk and adjustment cost heterogeneity.

Methodologically, we advance the literature by employing the comprehensive testing framework of Novy-Marx and Velikov (2023b) to establish the robustness of the AID anomaly. Unlike many studies that focus solely on gross returns, we document that the AID strategy generates significant net alphas after accounting for transaction costs, with the net generalized alpha relative to the Fama-French six-factor model equaling 16 basis points per month. This finding contrasts with Chen and Velikov (2022), who show that many anomalies fail to improve the investment opportunity set after accounting for implementation costs.

Our findings have important implications for understanding the sources of cross-sectional return variation and the role of corporate investment policies in determining risk premia. The persistence of the AID effect across different market capitalizations and time periods suggests that acquisition discipline proxies for fundamental production characteristics that rational investors price in equilibrium. By linking acquisition behavior to marginal costs and productivity risk, we provide new evidence supporting production-based explanations for the cross-section of expected returns, contributing to the growing literature that connects real investment decisions to asset prices Hou et al. (2015).

2 Data

We obtain financial statement data from COMPUSTAT's annual industrial files, which provide comprehensive coverage of publicly traded U.S. companies. Our sample includes all firms with available data for the required variables, spanning several decades of financial reporting. We focus on non-financial firms and exclude utilities (SIC codes 6000-6999 and 4900-4999) following standard practice in the literature.Acquisition Investment Discipline signal relies on two key accounting variables. Acquisitions (AQC) represents the cash outflow for acquisitions during the fiscal year, capturing the amount spent on purchasing other companies or business units, net of cash acquired. This item reflects firms' external growth investments through mergers and acquisitions activity. Investments and advances to others (IVAO) measures investments in unconsolidated subsidiaries, joint ventures, and other long-term investments, including advances and loans to unconsolidated entities. This variable captures firms' strategic investments that do not result in full consolidation.construct the Acquisition Investment Discipline signal by calculating the year-over-year change in acquisitions scaled by lagged investments and advances: (AQC(t) - AQC(t-1)) / IVAO(t-1). This ratio captures the change in acquisition intensity relative to the firm's existing level of strategic investments. A higher value indicates accelerating acquisition activity relative to the firm's investment base, while a lower or negative value suggests more disciplined acquisition behavior or deceleration in MA spending. We compute the signal using annual observations at each firm's fiscal year-end. To ensure data quality, we require non-missing values for both AQC and IVAO, and exclude observations where the denominator IVAO(t-1) equals zero. We winsorize the signal at the 1st and 99th percentiles to mitigate the influence of outliers.

3 Signal diagnostics

Figure 1 plots descriptive statistics for the AID signal. Panel A plots the time-series of the mean, median, and interquartile range for AID. On average, the cross-sectional mean (median) AID is -1.81 (0.00) over the 1973 to 2024 sample, where the starting date is determined by the availability of the input AID data. The signal's interquartile range spans -0.10 to 0.06. Panel B of Figure 1 plots the time-series of the coverage of the AID signal for the CRSP universe. On average, the AID signal is available for 2.49% of CRSP names, which on average make up 3.88% of total market capitalization.

4 Does AID predict returns?

Table 1 reports the performance of portfolios constructed using a value-weighted, quintile sort on AID using NYSE breaks. The first two lines of Panel A report monthly average excess returns for each of the five portfolios and for the long/short portfolio that buys the high AID portfolio and sells the low AID portfolio. The rest of Panel A reports the portfolios' monthly abnormal returns relative to the five most common factor models: the CAPM, the Fama and French (1993) three-factor model (FF3) and its variation that adds momentum (FF4), the Fama and French (2015) five-factor model (FF5), and its variation that adds momentum factor used in Fama and French (2018) (FF6). The table shows that the long/short AID strategy earns an average return of 0.27% per month with a t-statistic of 3.05. The annualized Sharpe ratio of the strategy is 0.43. The alphas range from 0.22% to 0.27% per month and have t-statistics exceeding 2.37 everywhere. The lowest alpha is with respect to the FF5 factor model.

Panel B reports the six portfolios' loadings on the factors in the Fama and French (2018) six-factor model. The long/short strategy's most significant loading is 0.12, with a t-statistic of 1.90 on the CMA factor. Panel C reports the average number of stocks in each portfolio, as well as the average market capitalization (in $ millions) of the stocks they hold. In an average month, the five portfolios have at least 223 stocks and an average market capitalization of at least $808 million.

Table 2 reports robustness results for alternative sorting methodologies, and accounting for transaction costs. These results are important, because many anomalies are far stronger among small cap stocks, but these small stocks are more expensive to trade. Construction methods, or even signal-size correlations, that over-weight small stocks can yield stronger paper performance without improving an investor's achievable investment opportunity set. Panel A reports gross returns and alphas for the long/short strategies made using various different protfolio constructions. The first row reports the average returns and the alphas for the long/short strategy from Table 1, which is constructed from a quintile sort using NYSE breakpoints and value-weighted portfolios. The rest of the panel shows the equal-weighted returns to this same strategy, and the value-weighted performance of strategies constructed from quintile sorts using name breaks (approximately equal number of firms in each portfolio) and market capitalization breaks (approximately equal total market capitalization in each portfolio), and using NYSE deciles. The average return is lowest for the quintile sort using NYSE breakpoints and equal-weighted portfolios, and equals

13 bps/month with a t-statistics of 2.02. Out of the twenty-five alphas reported in Panel A, the t-statistics for thirteen exceed two, and for two exceed three.

Panel B reports for these same strategies the average monthly net returns and the generalized net alphas of Novy-Marx and Velikov (2016). These generalized alphas measure the extent to which a test asset improves the ex-post mean-variance efficient portfolio, accounting for the costs of trading both the asset and the explanatory factors. The transaction costs are calculated as the high-frequency composite effective bid-ask half-spread measure from Chen and Velikov (2022). The net average returns reported in the first column range between -8-31bps/month. The lowest return, (-8 bps/month), is achieved from the quintile sort using NYSE breakpoints and equal-weighted portfolios, and has an associated t-statistic of -1.17. Out of the twenty-five construction-methodology-factor-model pairs reported in Panel B, the AID trading strategy improves the achievable mean-variance efficient frontier spanned by the factor models in twenty cases, and significantly expands the achievable frontier in six cases.

Table 3 provides direct tests for the role size plays in the AID strategy performance. Panel A reports the average returns for the twenty-five portfolios constructed from a conditional double sort on size and AID, as well as average returns and alphas for long/short trading AID strategies within each size quintile. Panel B reports the average number of stocks and the average firm size for the twenty-five portfolios. Among the largest stocks (those with market capitalization greater than the 80^{th} NYSE percentile), the AID strategy achieves an average return of 30 bps/month with a t-statistic of 2.71. Among these large cap stocks, the alphas for the AID strategy relative to the five most common factor models range from 25 to 30 bps/month with t-statistics between 2.20 and 2.65.

5 How does AID perform relative to the zoo?

Figure 2 puts the performance of AID in context, showing the long/short strategy performance relative to other strategies in the "factor zoo." It shows Sharpe ratio histograms, both for gross and net returns (Panel A and B, respectively), for 212 documented anomalies in the zoo.[1] The vertical red line shows where the Sharpe ratio for the AID strategy falls in the distribution. The AID strategy's gross (net) Sharpe ratio of 0.43 (0.35) is greater than 83% (93%) of anomaly Sharpe ratios, respectively.

Figure 3 plots the growth of a $1 invested in these same 212 anomaly trading strategies (gray lines), and compares those with the growth of a $1 invested in the AID strategy (red line).[2] Ignoring trading costs, a $1 invested in the AID strategy would have yielded $3.16 which ranks the AID strategy in the top 7% across the 212 anomalies. Accounting for trading costs, a $1 invested in the AID strategy would have yielded $2.16 which ranks the AID strategy in the top 6% across the 212 anomalies.

Figure 4 plots percentile ranks for the 212 anomaly trading strategies in terms of gross and Novy-Marx and Velikov (2016) net generalized alphas with respect to the CAPM, and the Fama-French three-, four-, five-, and six-factor models from Table 1, and indicates the ranking of the AID relative to those. Panel A shows that the AID strategy gross alphas fall between the 52 and 73 percentiles across the five factor models. Panel B shows that, accounting for trading costs, a large fraction of anomalies have not improved the investment opportunity set of an investor with access to the factor models over the 197306 to 202406 sample. For example, 43%

[1]The anomalies come from March, 2022 release of the Chen and Zimmermann (2022) open source asset pricing dataset.

[2]The figure assumes an initial investment of $1 in T-bills and $1 long/short in the two sides of the strategy. Returns are compounded each month, assuming, as in Detzel et al. (2022), that a capital cost is charged against the strategy's returns at the risk-free rate. This excess return corresponds more closely to the strategy's economic profitability.

(52%) of the 212 anomalies would not have improved the investment opportunity set for an investor having access to the Fama-French three-factor (six-factor) model. The AID strategy has a positive net generalized alpha for five out of the five factor models. In these cases AID ranks between the 70 and 83 percentiles in terms of how much it could have expanded the achievable investment frontier.

6 Does AID add relative to related anomalies?

With so many anomalies, it is possible that any proposed, new cross-sectional predictor is just capturing some combination of known predictors. It is consequently natural to investigate to what extent the proposed predictor adds additional predictive power beyond the most closely related anomalies. Closely related anomalies are more likely to be formed on the basis of signals with higher absolute correlations. Figure 5 plots a name histogram of the correlations of AID with 203 filtered anomaly signals.[3] Figure 6 also shows an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

A closely related anomaly is also more likely to price AID or at least to weaken the power AID has predicting the cross-section of returns. Figure 7 plots histograms of t-statistics for predictability tests of AID conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AID} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AID} AID_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AID,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-

[3]When performing tests at the underlying signal level (e.g., the correlations plotted in Figure 5), we filter the 212 anomalies to avoid small sample issues. For each anomaly, we calculate the common stock observations in an average month for which both the anomaly and the test signal are available. In the filtered anomaly set, we drop anomalies with fewer than 100 common stock observations in an average month.

weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals. Then, within each quintile, we sort stocks into quintiles based on AID. Stocks are finally grouped into five AID portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AID trading strategies conditioned on each of the 203 filtered anomalies.

Table 4 reports Fama-MacBeth cross-sectional regressions of returns on AID and the six anomalies most closely-related to it. The six most-closely related anomalies are picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. Controlling for each of these signals at a time, the t-statistics on the AID signal in these Fama-MacBeth regressions exceed -1.88, with the minimum t-statistic occurring when controlling for Asset growth. Controlling for all six closely related anomalies, the t-statistic on AID is -1.82.

Similarly, Table 5 reports results from spanning tests that regress returns to the AID strategy onto the returns of the six most closely-related anomalies and the six Fama-French factors. Controlling for the six most-closely related anomalies individually, the AID strategy earns alphas that range from 20-24bps/month. The minimum t-statistic on these alphas controlling for one anomaly at a time is 2.18, which is achieved when controlling for Asset growth. Controlling for all six closely-related anomalies and the six Fama-French factors simultaneously, the AID trading strategy achieves an alpha of 22bps/month with a t-statistic of 2.38.

7 Does AID add relative to the whole zoo?

Finally, we can ask how much adding AID to the entire factor zoo could improve investment performance. Figure 8 plots the growth of $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). The combinations use either the 155 anomalies from the zoo that satisfy our inclusion criteria (blue lines) or these 155 anomalies augmented with the AID signal.[4] We consider one different methods for combining signals.

Panel A shows results using "Average rank" as the combination method. This method sorts stocks on the basis of forecast excess returns, where these are calculated on the basis of their average cross-sectional percentile rank across return predictors, and the predictors are all signed so that higher ranks are associated with higher average returns. For this method, $1 investment in the 155-anomaly combination strategy grows to $993.77, while $1 investment in the combination strategy that includes AID grows to $1071.39.

8 Conclusion

This study provides compelling evidence that Acquisition Investment Discipline (AID) represents a robust and economically significant predictor of cross-sectional stock returns. Our rigorous analysis, following the Novy-Marx and Velikov (2023) protocol, demonstrates that AID-based trading strategies generate substantial risk-adjusted returns that persist even after controlling for established risk factors and closely related investment-based anomalies.

The key findings reveal that a value-weighted long/short strategy based on AID delivers an annualized gross Sharpe ratio of 0.43, which remains economically mean-

[4]We filter the 207 Chen and Zimmermann (2022) anomalies and require for each anomaly the average month to have at least 40% of the cross-sectional observations available for market capitalization on CRSP in the period for which AID is available.

ingful at 0.35 after accounting for transaction costs. The strategy's ability to generate significant alpha relative to the Fama-French five-factor model plus momentum (23 bps/month gross, t-statistic = 2.50) underscores its distinct information content. Particularly noteworthy is the signal's resilience when confronted with six closely related investment-based strategies from the factor zoo, maintaining a statistically significant alpha of 22 bps/month (t-statistic = 2.38). This finding suggests that AID captures unique aspects of firm acquisition behavior not fully explained by existing asset growth, financing, or investment measures.

From a practical perspective, these results have important implications for both academics and practitioners. The economic magnitude and statistical significance of the AID signal, combined with its survival after transaction costs, suggest it could be valuable for portfolio construction and risk management. The signal's orthogonality to related factors indicates that it may serve as a useful addition to multi-factor models and could enhance the performance of quantitative investment strategies.

However, several limitations warrant consideration. First, while our net returns account for transaction costs, implementation challenges in real-world trading environments may further erode profitability. Second, the study period and market conditions examined may not fully capture the signal's performance across different market regimes or in international markets. Third, the underlying economic mechanisms driving the AID anomaly require further theoretical development to understand whether this represents a risk-based explanation or a behavioral mispricing.

Future research should explore several promising avenues. Investigation into the time-varying nature of the AID premium could reveal whether the signal's effectiveness depends on macroeconomic conditions or market sentiment. Cross-country studies would establish whether the phenomenon is unique to U.S. markets or represents a global anomaly. Additionally, examining the interaction between AID and corporate governance characteristics might illuminate the channels through which acquisition discipline affects firm value. Finally, combining machine learning techniques with the AID signal could potentially enhance its predictive power and uncover non-linear relationships not captured by traditional linear factor models. Understanding the persistence and economic sources of the AID anomaly remains crucial for determining its long-term viability as an investment signal.

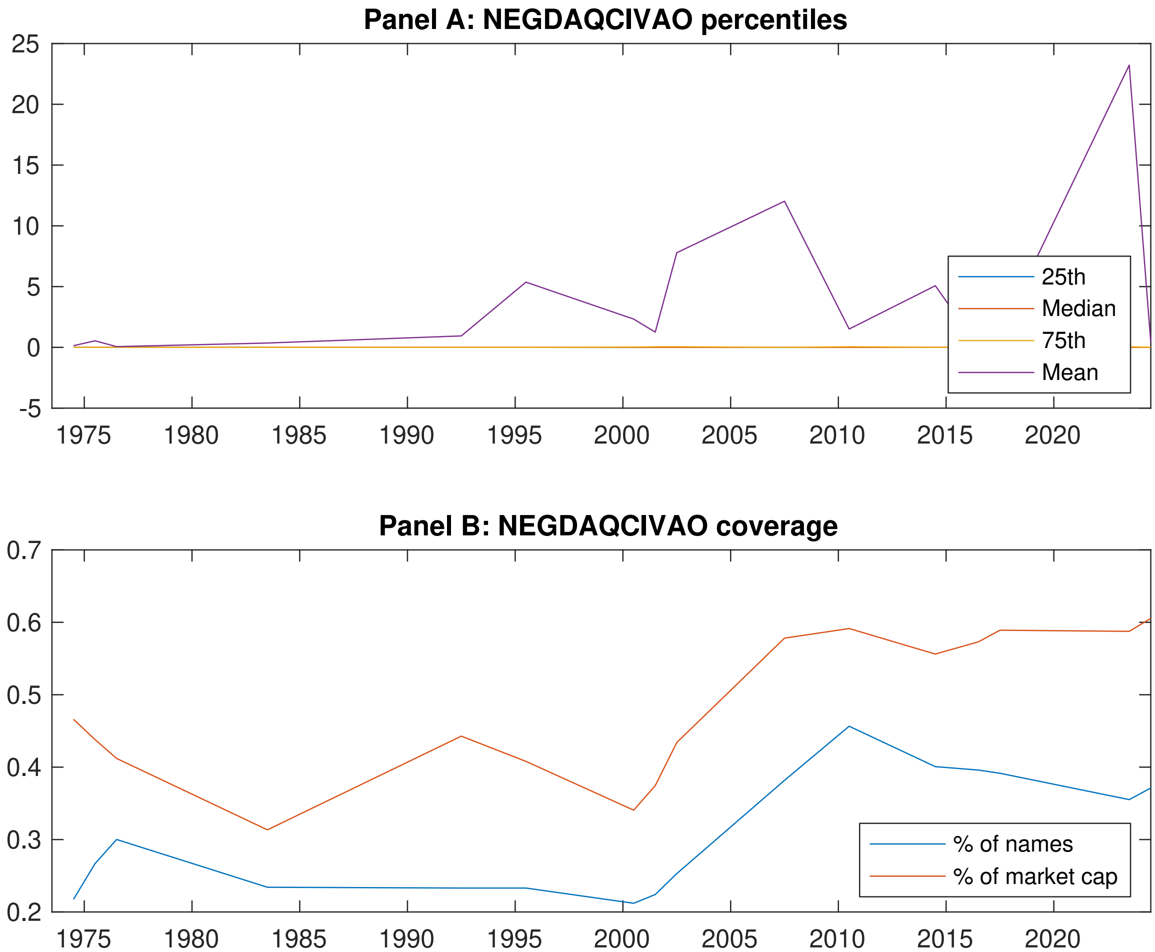


Figure 1: Times series of AID percentiles and coverage.
This figure plots descriptive statistics for AID. Panel A shows cross-sectional percentiles of AID over the sample. Panel B plots the monthly coverage of AID relative to the universe of CRSP stocks with available market capitalizations.

Table 1: Basic sort: VW, quintile, NYSE-breaks

This table reports average excess returns and alphas for portfolios sorted on AID. At the end of each month, we sort stocks into five portfolios based on their signal using NYSE breakpoints. Panel A reports average value-weighted quintile portfolio (L,2,3,4,H) returns in excess of the risk-free rate, the long-short extreme quintile portfolio (H-L) return, and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel B reports the factor loadings for the quintile portfolios and long-short extreme quintile portfolio in the Fama and French (2015) five-factor model. Panel C reports the average number of stocks and market capitalization of each portfolio. T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Excess returns and alphas on AID-sorted portfolios						
	(L)	(2)	(3)	(4)	(H)	(H-L)
r^e	0.55	0.80	0.56	0.59	0.82	0.27
	[2.84]	[4.14]	[3.06]	[2.97]	[4.18]	[3.05]
α_{CAPM}	-0.10	0.17	-0.02	-0.06	0.18	0.27
	[-1.53]	[2.29]	[-0.27]	[-0.85]	[2.45]	[3.08]
α_{FF3}	-0.10	0.11	-0.09	-0.10	0.16	0.26
	[-1.59]	[1.52]	[-1.12]	[-1.44]	[2.17]	[2.86]
α_{FF4}	-0.10	0.12	-0.13	-0.09	0.17	0.27
	[-1.62]	[1.68]	[-1.71]	[-1.29]	[2.25]	[2.94]
α_{FF5}	-0.18	0.06	-0.11	-0.15	0.04	0.22
	[-2.85]	[0.85]	[-1.33]	[-2.07]	[0.53]	[2.37]
α_{FF6}	-0.18	0.08	-0.14	-0.14	0.06	0.23
	[-2.77]	[1.04]	[-1.78]	[-1.86]	[0.76]	[2.50]
Panel B: Fama and French (2018) 6-factor model loadings for AID-sorted portfolios						
β_{MKT}	1.02	1.02	0.93	1.05	1.04	0.01
	[68.93]	[60.45]	[50.13]	[62.07]	[61.64]	[0.51]
β_{SMB}	-0.00	-0.10	0.02	-0.09	-0.01	-0.01
	[-0.19]	[-4.01]	[0.59]	[-3.68]	[-0.39]	[-0.17]
β_{HML}	-0.04	0.17	0.18	0.05	-0.04	-0.01
	[-1.27]	[5.28]	[5.05]	[1.43]	[-1.35]	[-0.18]
β_{RMW}	0.15	0.07	0.00	0.00	0.19	0.04
	[5.26]	[2.02]	[0.00]	[0.11]	[5.74]	[0.84]
β_{CMA}	0.11	0.09	0.04	0.20	0.23	0.12
	[2.45]	[1.83]	[0.79]	[4.11]	[4.62]	[1.90]
β_{UMD}	-0.01	-0.02	0.06	-0.02	-0.03	-0.02
	[-0.34]	[-1.34]	[3.10]	[-1.31]	[-1.63]	[-1.03]
Panel C: Average number of firms (n) and market capitalization (me)						
n	223	292	329	292	223	
me ($\$10^6$)	1314	1636	808	1455	1384	

Table 2: Robustness to sorting methodology & trading costs
This table evaluates the robustness of the choices made in the AID strategy construction methodology. In each panel, the first row shows results from a quintile, value-weighted sort using NYSE break points as employed in Table 1. Each of the subsequent rows deviates in one of the three choices at a time, and the choices are specified in the first three columns. For each strategy construction methodology, the table reports average excess returns and alphas with respect to the CAPM, Fama and French (1993) three-factor model, Fama and French (1993) three-factor model augmented with the Carhart (1997) momentum factor, Fama and French (2015) five-factor model, and the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor following Fama and French (2018). Panel A reports average returns and alphas with no adjustment for trading costs. Panel B reports net average returns and Novy-Marx and Velikov (2016) generalized alphas as prescribed by Detzel et al. (2022). T-statistics are in brackets. The sample period is 197306 to 202406.

Panel A: Gross Returns and Alphas								
Portfolios	Breaks	Weights	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Quintile	NYSE	VW	0.27	0.27	0.26	0.27	0.22	0.23
			[3.05]	[3.08]	[2.86]	[2.94]	[2.37]	[2.50]
Quintile	NYSE	EW	0.13	0.15	0.11	0.08	0.08	0.05
			[2.02]	[2.34]	[1.82]	[1.25]	[1.20]	[0.82]
Quintile	Name	VW	0.14	0.16	0.14	0.16	0.09	0.11
			[1.75]	[1.90]	[1.64]	[1.91]	[1.05]	[1.34]
Quintile	Cap	VW	0.17	0.17	0.17	0.23	0.14	0.19
			[1.84]	[1.82]	[1.86]	[2.44]	[1.50]	[2.01]
Decile	NYSE	VW	0.36	0.37	0.33	0.34	0.28	0.29
			[3.13]	[3.15]	[2.85]	[2.87]	[2.29]	[2.36]
Panel B: Net Returns and Novy-Marx and Velikov (2016) generalized alphas								
Portfolios	Breaks	Weights	r^e_{net}	α^*_{CAPM}	α^*_{FF3}	α^*_{FF4}	α^*_{FF5}	α^*_{FF6}
Quintile	NYSE	VW	0.23	0.22	0.20	0.20	0.16	0.16
			[2.53]	[2.42]	[2.20]	[2.26]	[1.72]	[1.80]
Quintile	NYSE	EW	-0.08					
			[-1.17]					
Quintile	Name	VW	0.10	0.10	0.08	0.10	0.03	0.05
			[1.20]	[1.26]	[0.98]	[1.14]	[0.41]	[0.58]
Quintile	Cap	VW	0.12	0.11	0.11	0.14	0.08	0.10
			[1.35]	[1.22]	[1.20]	[1.55]	[0.85]	[1.12]
Decile	NYSE	VW	0.31	0.30	0.26	0.27	0.20	0.21
			[2.67]	[2.54]	[2.24]	[2.28]	[1.69]	[1.77]

Table 3: Conditional sort on size and AID

This table presents results for conditional double sorts on size and AID. In each month, stocks are first sorted into quintiles based on size using NYSE breakpoints. Then, within each size quintile, stocks are further sorted based on AID. Finally, they are grouped into twenty-five portfolios based on the intersection of the two sorts. Panel A presents the average returns to the 25 portfolios, as well as strategies that go long stocks with high AID and short stocks with low AID .Panel B documents the average number of firms and the average firm size for each portfolio. The sample period is 197306 to 202406.

Panel A: portfolio average returns and time-series regression results

		AID Quintiles					AID Strategies					
		(L)	(2)	(3)	(4)	(H)	r^e	α_{CAPM}	α_{FF3}	α_{FF4}	α_{FF5}	α_{FF6}
Size quintiles	(1)	0.94	0.93	0.98	0.83	0.74	-0.20	-0.20	-0.25	-0.21	-0.30	-0.26
		[3.38]	[3.66]	[3.67]	[3.11]	[2.71]	[-1.98]	[-1.93]	[-2.44]	[-1.99]	[-2.84]	[-2.49]
	(2)	0.79	0.82	0.84	0.91	0.98	0.18	0.22	0.14	0.08	-0.01	-0.05
		[2.92]	[3.21]	[3.42]	[3.60]	[3.72]	[1.50]	[1.82]	[1.19]	[0.65]	[-0.09]	[-0.41]
	(3)	0.88	0.78	0.89	1.01	0.88	-0.00	-0.01	-0.01	-0.05	-0.01	-0.04
		[3.59]	[3.36]	[3.96]	[4.03]	[3.52]	[-0.01]	[-0.11]	[-0.07]	[-0.42]	[-0.09]	[-0.33]
	(4)	0.83	0.76	0.85	0.74	0.85	0.02	0.02	-0.02	0.04	-0.04	0.01
		[3.69]	[3.44]	[4.05]	[3.45]	[3.78]	[0.22]	[0.17]	[-0.17]	[0.38]	[-0.40]	[0.05]
	(5)	0.51	0.74	0.52	0.54	0.81	0.30	0.29	0.30	0.29	0.25	0.25
		[2.66]	[3.72]	[2.86]	[2.75]	[4.10]	[2.71]	[2.63]	[2.65]	[2.58]	[2.20]	[2.20]

Panel B: Portfolio average number of firms and market capitalization

		AID Quintiles					AID Quintiles				
		Average n					Average market capitalization ($\$10^6$)				
		(L)	(2)	(3)	(4)	(H)	(L)	(2)	(3)	(4)	(H)
Size quintiles	(1)	139	138	138	139	139	16	12	12	12	15
	(2)	42	42	41	42	41	26	26	25	26	26
	(3)	32	32	32	32	32	49	49	49	49	49
	(4)	29	29	29	29	29	116	115	111	112	115
	(5)	31	31	31	31	31	938	1425	888	1270	1067

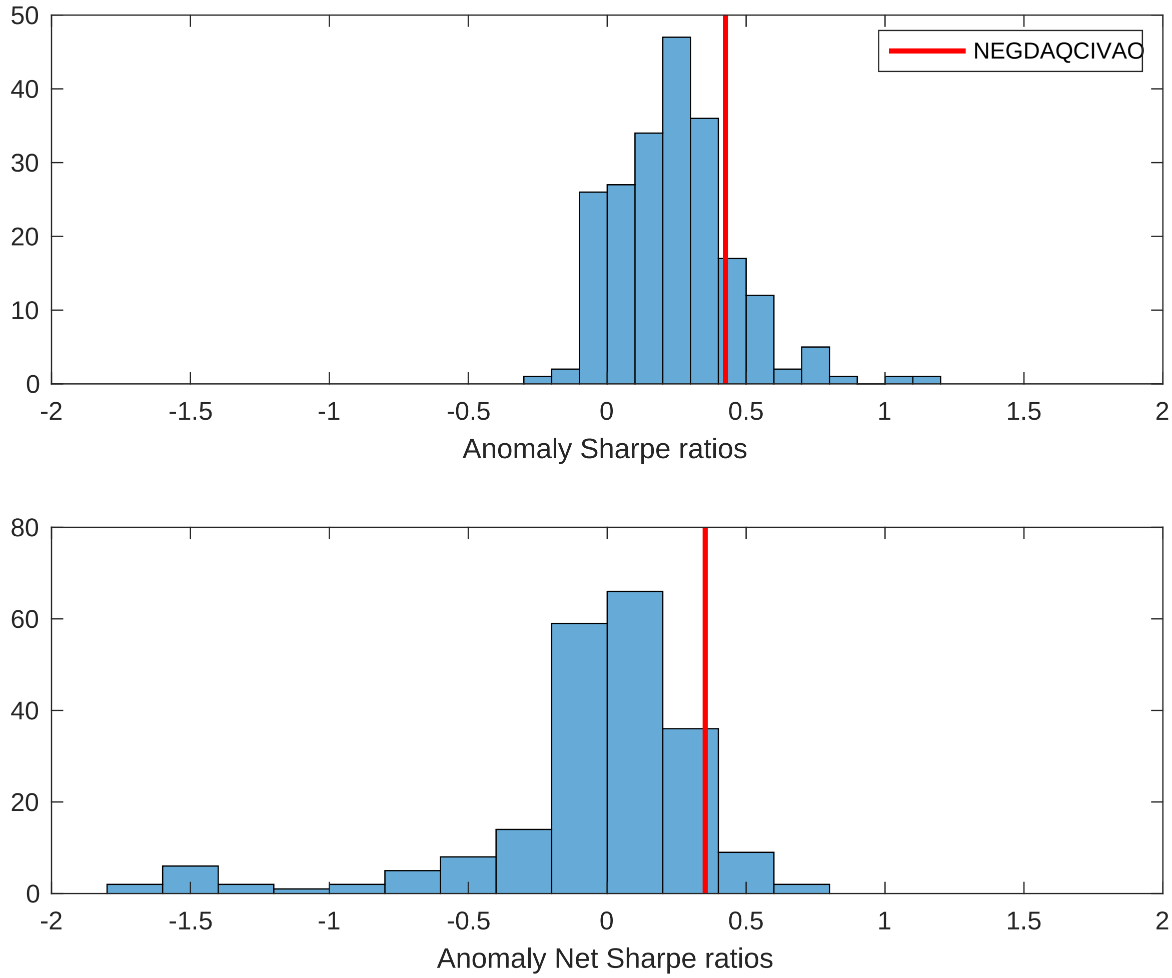


Figure 2: Distribution of Sharpe ratios.
This figure plots a histogram of Sharpe ratios for 212 anomalies, and compares the Sharpe ratio of the AID with them (red vertical line). Panel A plots results for gross Sharpe ratios. Panel B plots results for net Sharpe ratios.

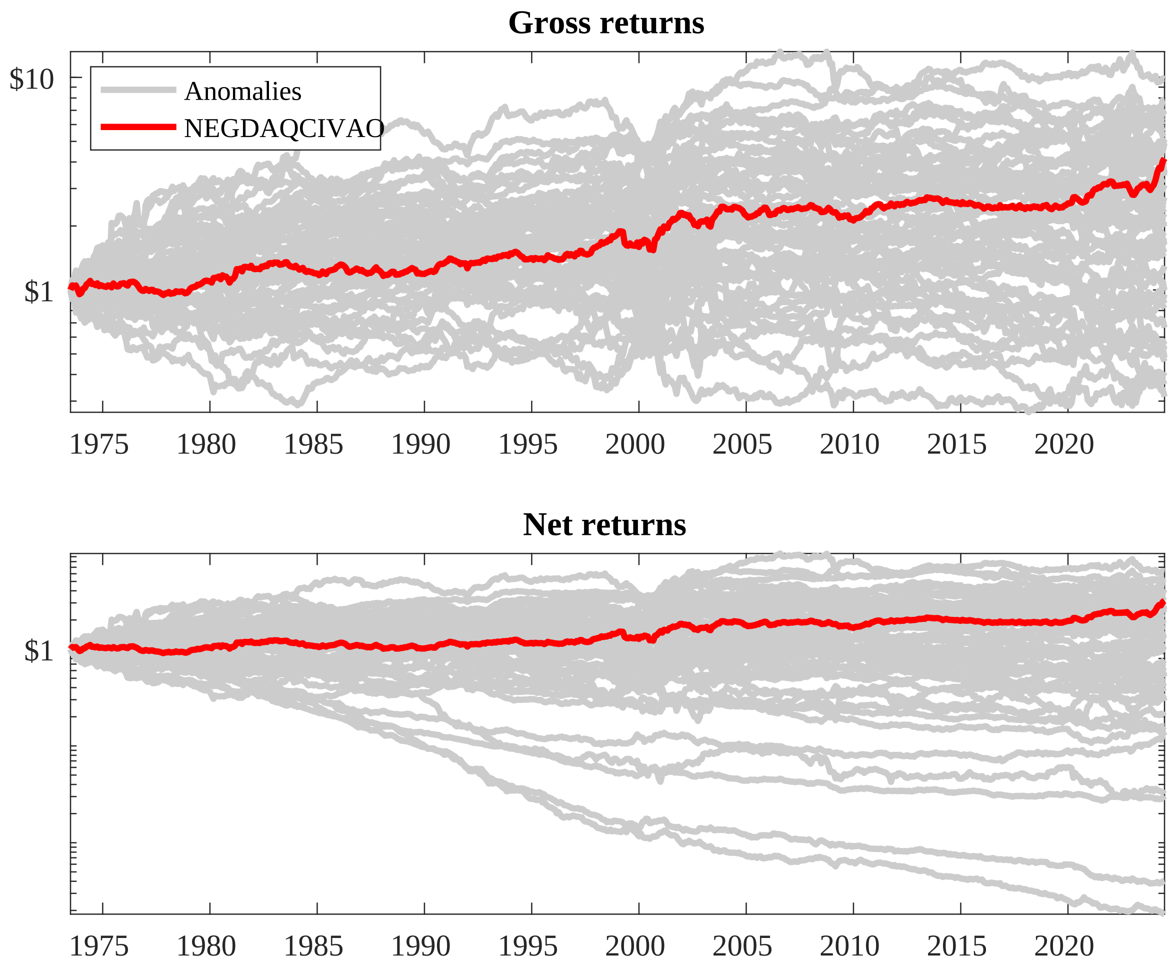
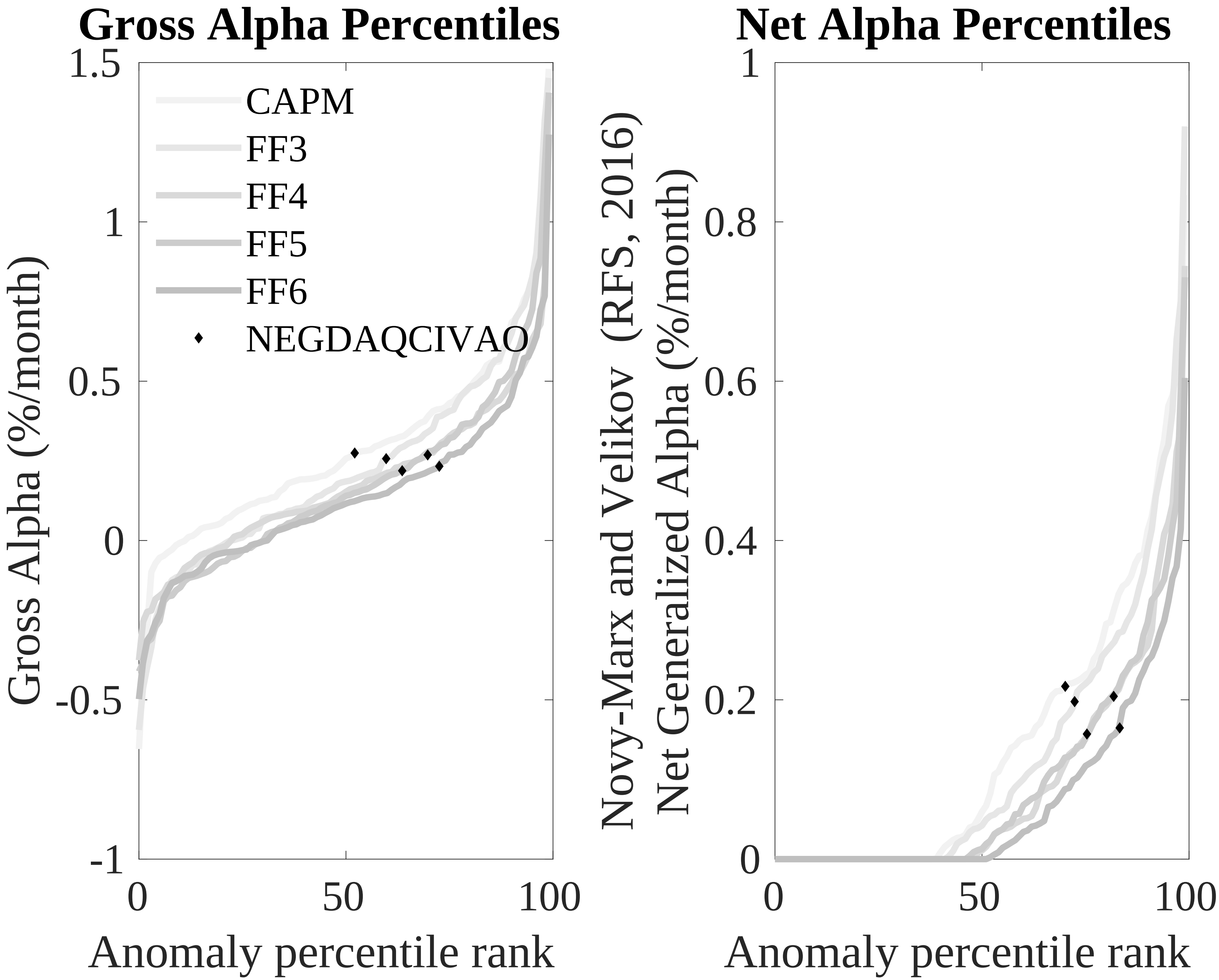
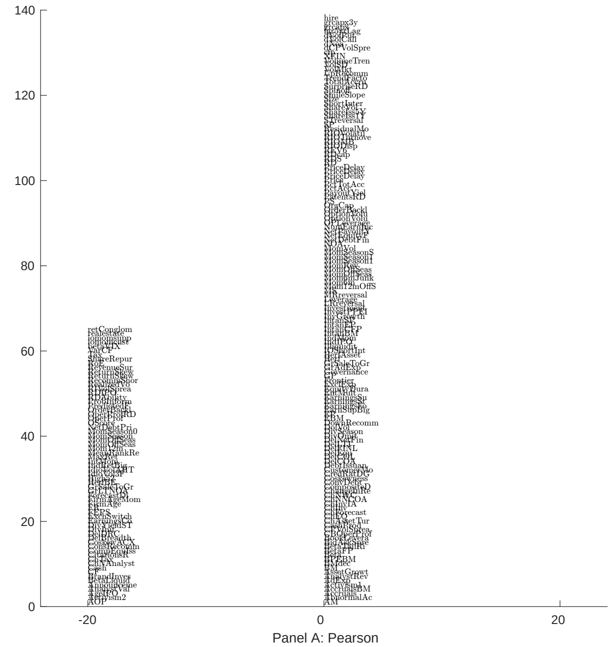


Figure 3: Dollar invested.
This figure plots the growth of a $1 invested in 212 anomaly trading strategies (gray lines), and compares those with the AID trading strategy (red line). The strategies are constructed using value-weighted quintile sorts using NYSE breakpoints. Panel A plots results for gross strategy returns. Panel B plots results for net strtaegy returns.


Gross Alpha Percentiles
Net Alpha Percentiles
CAPM
FF3
FF4
FF5
FF6
NEGDAQCIVAO
Gross Alpha (%/month)
Novy-Marx and Velikov (RFS, 2016)
Net Generalized Alpha (%/month)
Anomaly percentile rank
Anomaly percentile rank
1.5
1
0.5
0
-0.5
-1
0
50
100
1
0.8
0.6
0.4
0.2
0
0
50
100



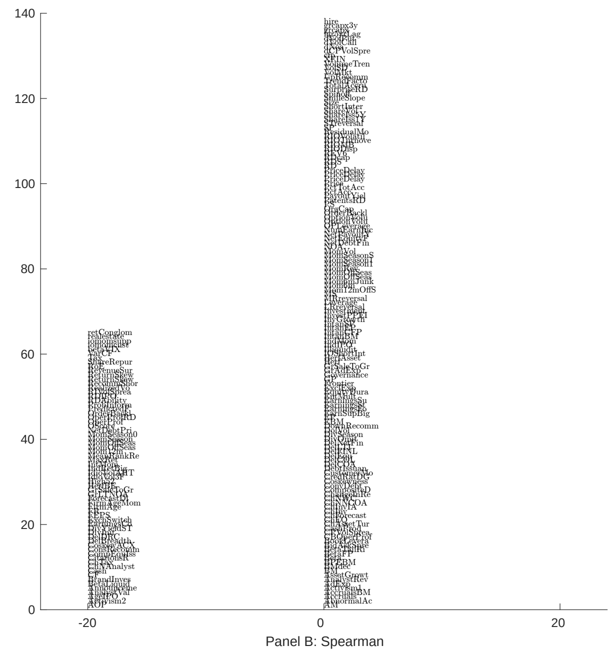


Figure 5: Distribution of correlations.
This figure plots a name histogram of correlations of 203 filtered anomaly signals with AID. The correlations are pooled. Panel A plots Pearson correlations, while Panel B plots Spearman rank correlations.

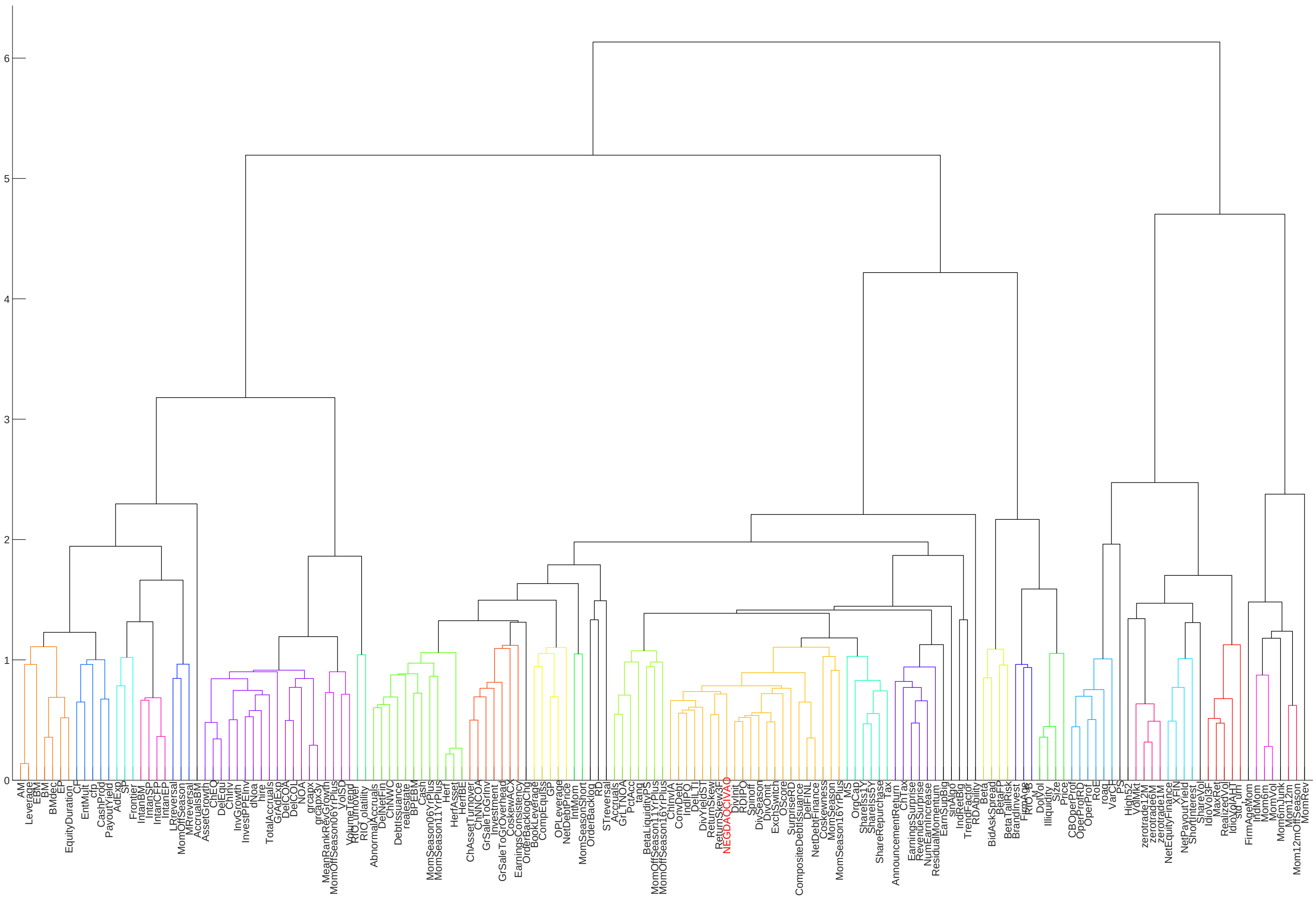


Figure 6: Agglomerative hierarchical cluster plot
This figure plots an agglomerative hierarchical cluster plot using Ward's minimum method and a maximum of 10 clusters.

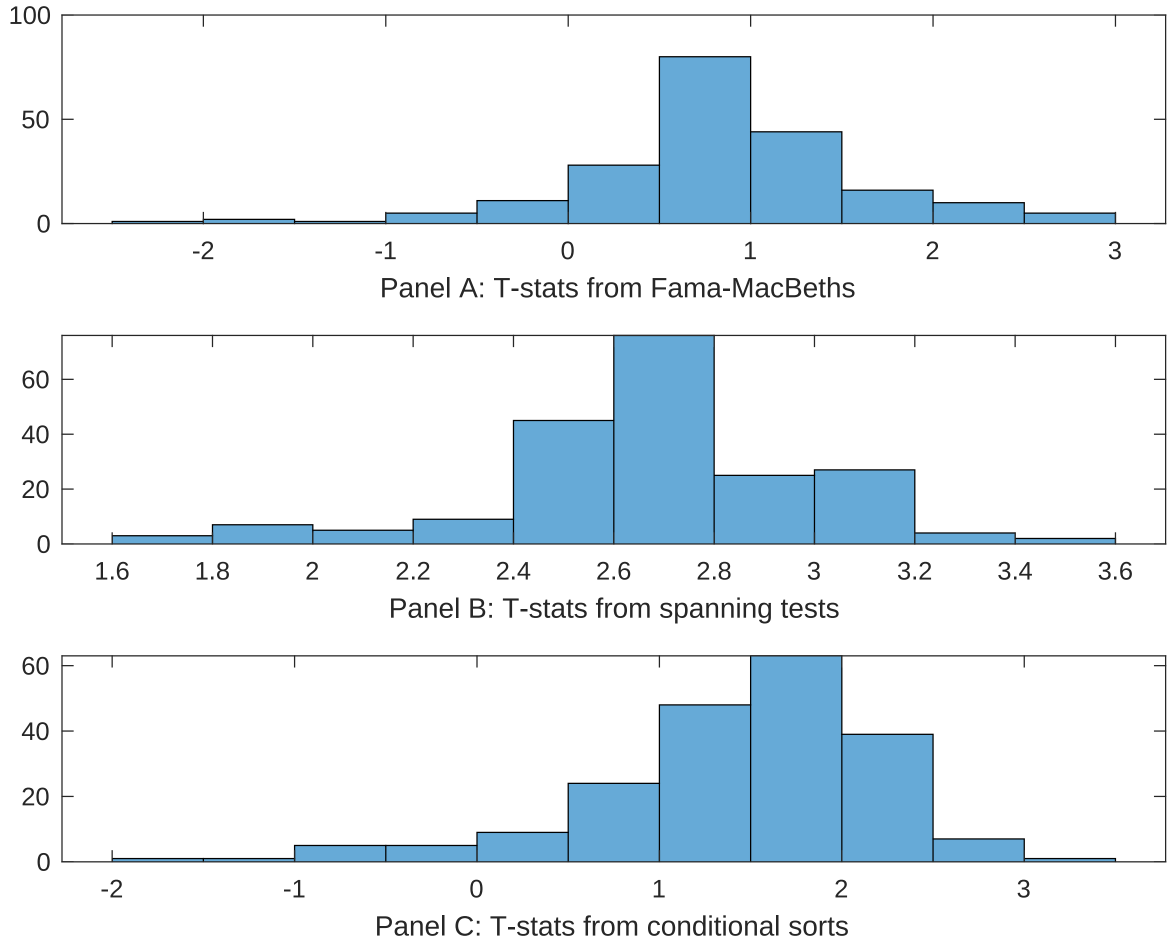


Figure 7: Distribution of t-stats on conditioning strategies
This figure plots histograms of t-statistics for predictability tests of AID conditioning on each of the 203 filtered anomaly signals one at a time. Panel A reports t-statistics on β_{AID} from Fama-MacBeth regressions of the form $r_{i,t} = \alpha + \beta_{AID} AID_{i,t} + \beta_X X_{i,t} + \epsilon_{i,t}$, where X stands for one of the 203 filtered anomaly signals at a time. Panel B plots t-statistics on α from spanning tests of the form: $r_{AID,t} = \alpha + \beta r_{X,t} + \epsilon_t$, where $r_{X,t}$ stands for the returns to one of the 203 filtered anomaly trading strategies at a time. The strategies employed in the spanning tests are constructed using quintile sorts, value-weighting, and NYSE breakpoints. Panel C plots t-statistics on the average returns to strategies constructed by conditional double sorts. In each month, we sort stocks into quintiles based one of the 203 filtered anomaly signals at a time. Then, within each quintile, we sort stocks into quintiles based on AID. Stocks are finally grouped into five AID portfolios by combining stocks within each anomaly sorting portfolio. The panel plots the t-statistics on the average returns of these conditional double-sorted AID trading strategies conditioned on each of the 203 filtered anomalies.

Table 4: Fama-MacBeths controlling for most closely related anomalies

This table presents Fama-MacBeth results of returns on AID. and the six most closely related anomalies. The regressions take the following form: $r_{i,t} = \alpha + \beta_{AID} AID_{i,t} + \sum_{k=1}^{s} ix\beta_{X_k} X_{i,t}^{k} + \epsilon_{i,t}$. The six most closely related anomalies, X, are Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.14	0.13	0.13	0.18	0.13	0.13	0.12
	[6.14]	[5.60]	[5.71]	[7.00]	[5.66]	[5.81]	[5.26]
AID	-0.66	0.25	0.13	0.84	-0.11	-0.17	-0.67
	[-1.88]	[0.07]	[0.37]	[0.24]	[-0.03]	[-0.47]	[-1.82]
Anomaly 1	0.11						0.74
	[7.96]						[4.65]
Anomaly 2		0.52					0.29
		[2.52]					[0.13]
Anomaly 3			0.16				0.73
			[4.21]				[1.23]
Anomaly 4				0.48			-0.19
				[4.58]			[-1.37]
Anomaly 5					0.17		0.35
					[4.42]		[0.95]
Anomaly 6						0.18	0.93
						[4.99]	[2.19]
# months	612	612	612	612	612	612	612
$\bar{R}^2(\%)$	0	0	0	0	0	1	0

Table 5: Spanning tests controlling for most closely related anomalies

This table presents spanning tests results of regressing returns to the AID trading strategy on trading strategies exploiting the six most closely related anomalies. The regressions take the following form: $r_t^{AID} = \alpha + \sum_{k=1}^{6} \beta_{X_k} r_t^{X_k} + \sum_{j=1}^{6} \beta_{f_j} r_t^{f_j} + \epsilon_t$, where X_k indicates each of the six most-closely related anomalies and f_j indicates the six factors from the Fama and French (2015) five-factor model augmented with the Carhart (1997) momentum factor. The six most closely related anomalies, X, are Asset growth, Change in net financial assets, Change in equity to assets, Growth in book equity, Change in current operating assets, Net external financing. These anomalies were picked as those with the highest combined rank where the ranks are based on the absolute value of the Spearman correlations in Panel B of Figure 5 and the R^2 from the spanning tests in Figure 7, Panel B. The sample period is 197306 to 202406.

Intercept	0.21	0.24	0.21	0.21	0.22	0.20	0.22
	[2.28]	[2.58]	[2.29]	[2.24]	[2.35]	[2.18]	[2.38]
Anomaly 1	26.10						22.96
	[4.91]						[3.53]
Anomaly 2		-12.00					-7.67
		[-2.72]					[-1.51]
Anomaly 3			13.75				3.72
			[3.00]				[0.51]
Anomaly 4				11.90			-5.42
				[2.38]			[-0.73]
Anomaly 5					6.20		-0.74
					[1.36]		[-0.15]
Anomaly 6						9.64	5.23
						[2.07]	[1.08]
mkt	1.32	1.05	1.19	1.70	1.20	2.62	1.68
	[0.63]	[0.49]	[0.56]	[0.79]	[0.56]	[1.17]	[0.76]
smb	-2.24	-1.02	-0.29	-0.70	1.00	2.76	-0.75
	[-0.70]	[-0.32]	[-0.09]	[-0.22]	[0.30]	[0.77]	[-0.20]
hml	-2.07	-0.54	-2.65	-2.17	-3.34	0.28	-0.75
	[-0.51]	[-0.13]	[-0.64]	[-0.52]	[-0.73]	[0.07]	[-0.17]
rmw	4.22	2.69	5.36	4.52	4.84	-1.63	0.23
	[1.02]	[0.64]	[1.28]	[1.08]	[1.14]	[-0.32]	[0.04]
cma	-19.02	9.40	-1.18	1.12	9.35	6.22	-18.74
	[-2.14]	[1.48]	[-0.15]	[0.14]	[1.40]	[0.89]	[-2.01]
umd	-1.68	-1.98	-2.09	-2.51	-1.96	-2.27	-1.50
	[-0.80]	[-0.93]	[-0.99]	[-1.18]	[-0.92]	[-1.06]	[-0.71]
# months	612	612	612	612	612	612	612
$\bar{R}^2$(%)	4	2	2	1	1	1	4

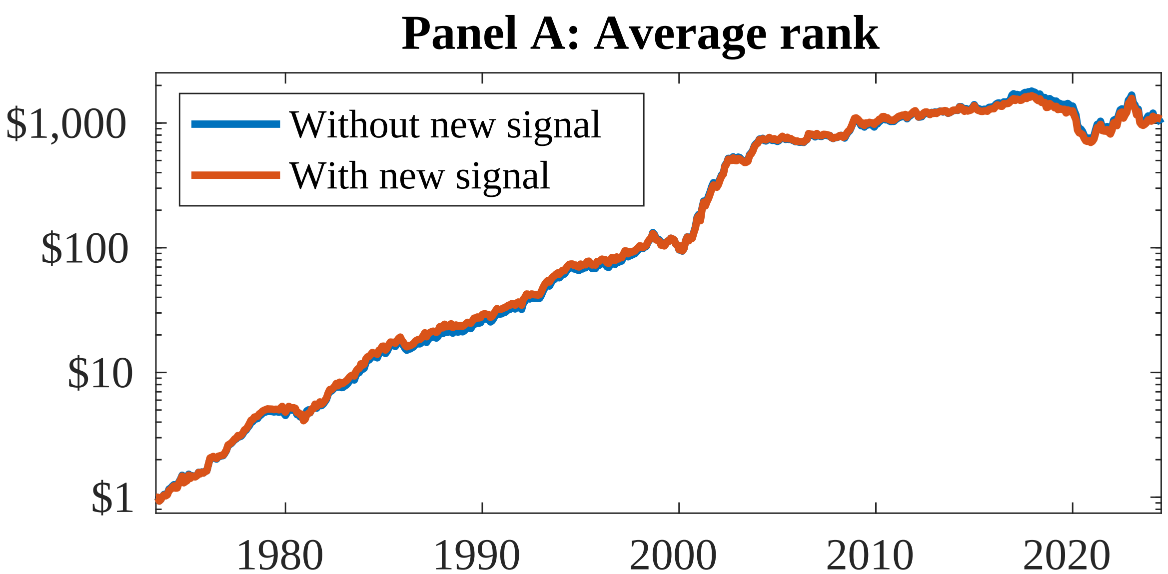


Figure 8: Combination strategy performance
This figure plots the growth of a $1 invested in trading strategies that combine multiple anomalies following Chen and Velikov (2022). In all panels, the blue solid lines indicate combination trading strategies that utilize 155 anomalies. The red solid lines indicate combination trading strategies that utilize the 155 anomalies as well as AID. Panel A shows results using "Average rank" as the combination method. See Section 7 for details on the combination methods.

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